

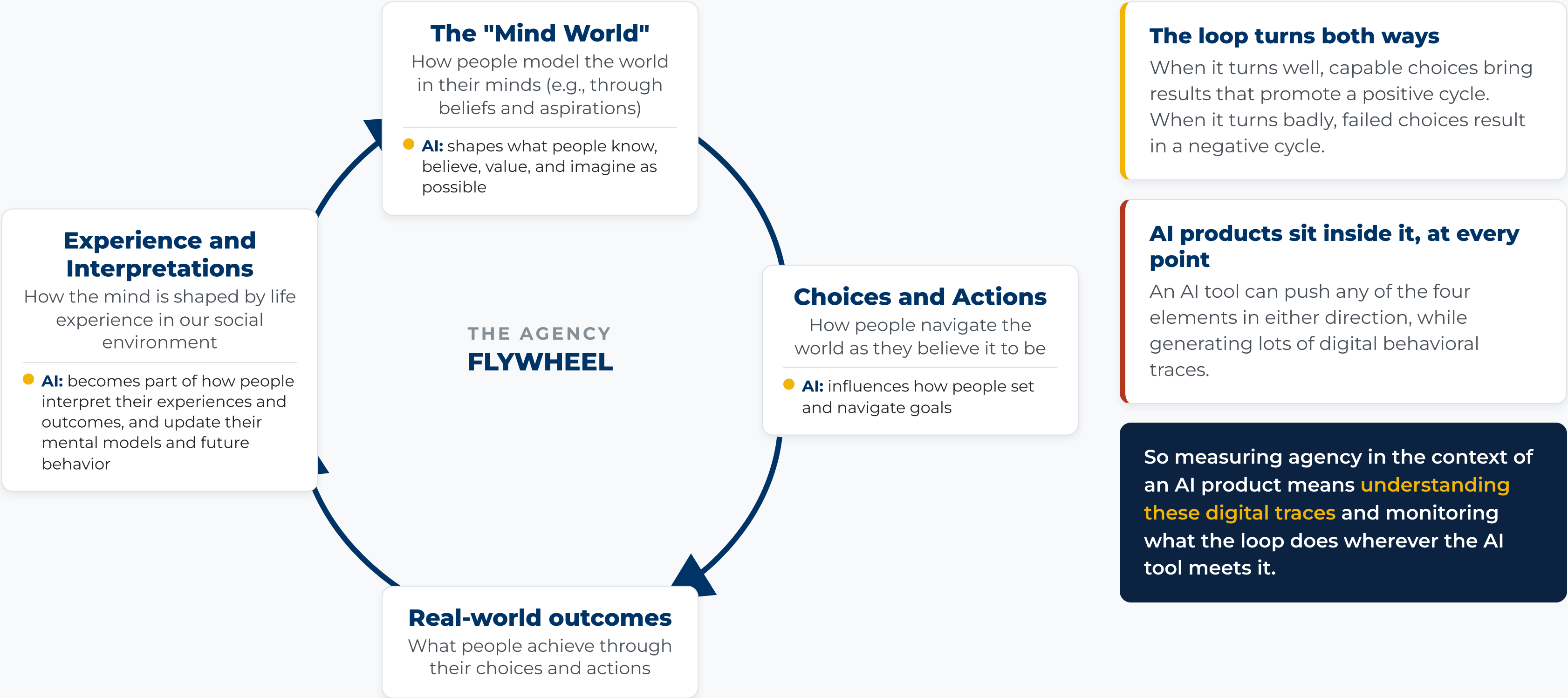
2026 NAIROBI ONSITE · TALK + WORKSHOP · 90 MINUTES

Measuring **agency** in an AI context

How can we tell when AI tool use is expanding or undermining human agency?
What is agency? Why should we measure it? How can we measure it to know
whether AI tools are actually supporting it?

Zezen Wu · Behavioral Scientist, The Agency Fund

Our worldview of agency in a flywheel



AI can **expand** agency, and it can hamper it

CAN EXPAND

THE FULL FINDING

People stay with tools that **make them feel more capable**, even when the tools make mistakes. AI can strengthen agency in five ways: instrumental, cognitive, affective, relational, and structural.

FLIP BACK

CAN HAMPER

THE FULL FINDING

A ChatGPT-style math tutor boosted practice scores by 48%, but students scored **17% below the control group** once the AI was removed. A guaranteed tutor that offered hints rather than solutions eliminated the harm, yet students using either tool remained unaware of the gap between feeling supported and actually learning.

FLIP BACK

CAN HAMPER

THE FULL FINDING

Across 1,375 participants, an accurate chatbot raised reasoning accuracy by 25 percentage points, but a faulty one drove it 15 points below unbiased performance. People followed wrong AI advice on **roughly four in five consulted trials** and became more confident even when misled; time pressure, incentives, and feedback reduced but did not eliminate the effect.

FLIP BACK

In all three studies, people looked engaged and felt capable while it happened. Usage and satisfaction cannot tell expansion of agency from its erosion, **which is why we measure agency directly.**

Sources: [How AI Is Affecting Human Agency](#) (Jigsaw, July 2025) · [Generative AI without guardrails can harm learning](#) (PNAS 122(26), 2025) · [Thinking, Fast, Slow, and Artificial: the rise of cognitive surrender](#) (Wharton working paper, 2026).

Agency needs to be defined to the context and measured as an outcome

It might be helpful to think of agency in terms of a theoretical framework. For example, in **social cognitive theory**, people are agents who intentionally shape their circumstances. Agency has four core elements in this framework:

1 • Intentionality

EXAMPLE • GOAL COMMITMENT

The learner states a business goal in her own words and **commits to a concrete next step** before the session ends.

FLIP BACK

2 • Forethought

EXAMPLE • PLANNING AND TRADEOFFS

The **weighs two ideas against each other**, asking what each would cost, earn, and risk in her local market.

FLIP BACK

3 • Self-reactiveness

EXAMPLE • COURSE CORRECTION

She returns with progress updates and **revises the plan** when the first supplier falls through, instead of abandoning the goal.

FLIP BACK

4 • Self-reflectiveness

EXAMPLE • CALIBRATED SELF-BELIEF

After the business pitch, she reviews what worked and what **didn't**, and updates her **mental models to develop the business plan**.

FLIP BACK

Think about **signals of agency** in two dimensions

Agency with respect to the end goals

Can users make progress toward outcomes they value, whether through, around, or beyond the tool?

Agency with respect to the digital tool

Can users understand, question, adapt, and control the tool, deciding when and how to rely on it?

Internally experienced agency

People's thoughts and feelings



ON-PLATFORM DATA · IN-APP MICRO-SURVEY

Short in-app surveys probing users' general sense of self-efficacy or perceived control over their ultimate goal over time (e.g., How confident is the user about developing a successful business idea? How much control do they feel over reaching their ultimate goal?)



OFF-PLATFORM DATA · PHONE OR IN-PERSON SURVEY

Longer phone/in-person surveys using validated agency measures that **systematically** probe users' general sense of self-efficacy or perceived control over their ultimate goal over time.



ON-PLATFORM DATA · IN-APP MICRO-SURVEY

Short in-app surveys probing whether users felt able to understand, question, or adopt an AI mentor's advice.



OFF-PLATFORM DATA · VALIDATED SCALES

Longer phone/in-person surveys using validated agency measures that systematically probe users' trust in AI and perceived dependence versus empowerment when using a tool.

Externally expressed agency

People's choices and actions



ON-PLATFORM DATA · CONVERSATION LOGS

Longitudinal user signals of agency in AI conversation logs, especially those related to setting and navigating longer-term goals or demonstrating resilience despite difficulties.



OFF-PLATFORM DATA · RUBRIC-RATED PITCHES

Longer phone/in-person surveys tracking how users develop their business ideas; rubric-rated observations of ideas and pitches; or classroom observation of teamwork, ideally linked to downstream outcomes like business earnings.



ON-PLATFORM DATA · DIGITAL TRACES

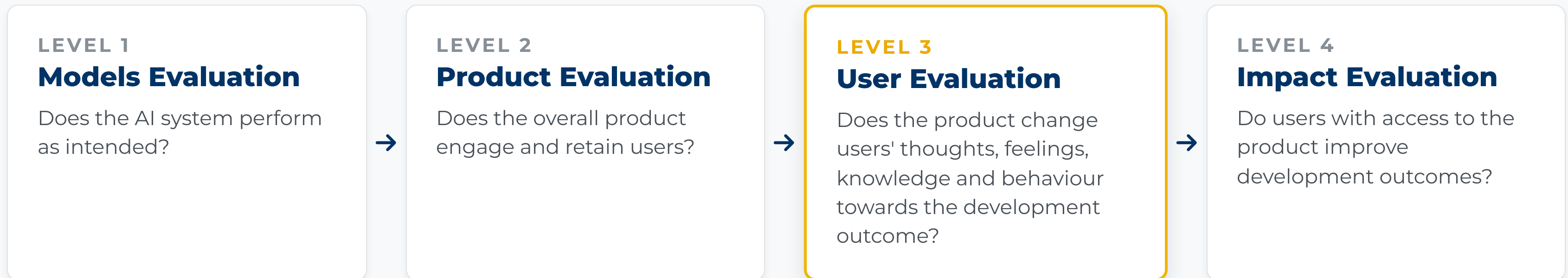
Digital traces (e.g., interaction patterns or conversation logs) showing whether users understand, question, or adopt an AI mentor's advice.



OFF-PLATFORM DATA · OBSERVATION AND INTERVIEWS

Observation or interview data (e.g., usability studies) on tool usage for developing business ideas.

Agency measurement sits at **Level 3**



Nobody can run an RCT on every product change. **Validated agency measures at Level 3** move when the product changes, and they speak to what Level 4 cares about. That is what lets teams anchor their A/B tests on something that matters.

HOW WE SUGGEST YOU DO IT • THREE STEPS THAT LOOP

Teams should specify what agency means for their specific users and contexts

1

Define agency for your local problem

What kind of agency is the tool supposed to support? **For whom? To what end?** Anchor the definition in your program's theory of change.



2

Measure with the right tools, on the right data, at the right moment

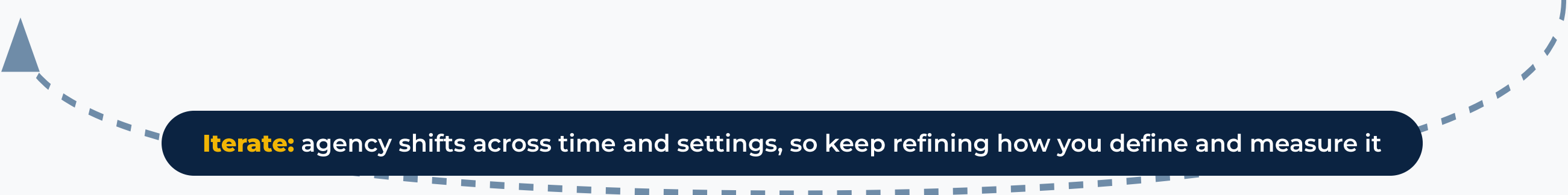
Combine **on-platform** signals, such as in-app surveys and digital traces, with **off-platform** surveys, interviews, observations, and records.



3

Test and validate your measurement tools iteratively

Check **reliability and validity** for your context, and compare LLM-based measures against a human-coded gold standard.



Iterate: agency shifts across time and settings, so keep refining how you define and measure it

PART 2 · WORKSHOP

Now you find the agency, **line by line.**

Why conversations? Because they can hold rich signals of agency, especially when a human is talking with an AI. You will read 18 synthetic conversations across three domains, answer four questions and an open note, and compare your labels with an AI judge at the end.

ACTIVITY • ABOUT 40 MINUTES

Your turn: code conversations like a measurement team

1 Pick one card from a domain pile

Education, health, or agriculture: choose the world closest to your work. The card's QR code is your personal labeller link.

2 Open the link on your phone or laptop

You will see ten conversations between users and a fictional AI product, plus the four coding questions. Some are two lines, some are long: real traffic looks like that.

3 Code alone, top to bottom, about 20 minutes

Answer the four questions for each conversation and leave one open note. Four or five conversations is a good pace; nobody needs to finish ten.

4 Then compare labels at your table, about 15 minutes

Find one conversation where you disagreed. Argue it out: what evidence did each of you use? Nominate your hardest case for the debrief.

THREE ARCHETYPES, ALL FICTIONAL

AI educational mentor

Business-idea coaching inside a school entrepreneurship programme

AI health assistant

Post-visit follow-up line for clinic patients

AI agricultural advisor

Crop, weather and market Q&A for smallholder farmers

All synthetic. The products are fictional and every conversation was written for this exercise. No real user data, no real organization.

Four questions, one open note

1

BELIEFS AND ASPIRATIONS

Belief or aspiration shift
Does the user express a new or changed sense of what is possible for them? New hope counts ("never thought of it like that") and lost hope counts too ("business is for rich kids"). A long-held goal merely restated does not.

2

PURPOSEFUL ACTION AND CHOICES

Specific action or decision
Does the user state a specific action or decision they will take, or have already taken? Concrete plans and completed actions count. "Maybe" and "we will see" do not.

3

REAL-WORLD OUTCOMES

Links action to a life outcome
Does the user connect the actions or advice to a real-life outcome they care about: income, harvest, health, school fees? They must say it themselves; following instructions with no stated why is No.

4

INTERPRETED EXPERIENCE

Uses past experience
Does the user refer to a past success or failure and use it to inform what they do now? The interpretation can be wrong and still count: judge whether experience is used, not whether the conclusion is correct.

OPEN NOTE Anything else you notice, especially the loop turning backwards: growing discouragement, over-reliance, decisions handed to the AI. Short conversations with nothing to code are normal; all-No is a real answer.



SET

5 min

10 min

15 min

20 min

-1 min

+1 min

Start

Reset

Now we find out how much we disagreed

1 Human agreement, per domain

We open Calibrate's agreement view together: how often did coders give the same answer to the same question about the same conversation?

2 The LLM judge, same rubric

An AI evaluator has already run your exact four questions across all 18 conversations. We compare its verdicts with the room's.

3 The hard cases

Two or three tables tell us where they split and what evidence each side used. Those arguments are the raw material for a sharper rubric.

The point of the exercise: agency leaves signals in text, and we want to identify them **consistently**. Human coders set the standard. Once a rubric survives this contest with human judgment, an LLM can apply it across thousands of conversations. That pairing is what makes agency measurement scale.

Calibrate your expectations

When trained medical experts coded real conversations in our Noora Health pilot, they agreed with each other less than half the time. Disagreement is normal. It is also the material we work with.

This was Step 3 in miniature: humans first, the machine second, and the argument between them is where the measure improves.

Agency measurement is interesting, necessary, and complex. **Start anyway.**

The blog →

Measuring Agency in a Digital Context, by The Agency Fund and Google Jigsaw

The Claude skill →

ask-agency-measurement: a coach that turns these frameworks into a plan for your product

The NotebookLM →

Curated papers and validated agency scales you can interrogate conversationally

The evaluation playbook →

The full four-level framework for evaluating GenAI products, level by level